# Introduction to Programming

**Team Assignment | Telegram Bot**

May 11, 2026

# Telegram Bot for Programming Practice and File-Based Data Processing

The bot combines several programming concepts studied during the course: Telegram bot handlers, file processing, ZIP archives, dictionaries, pandas data analysis, simple calculations, timers, stickers, and location handling.

**The bot can:**
- process ZIP files
- search information inside files
- work with datasets
- create simple charts
- use calculator and timer functions
- send stickers and handle location

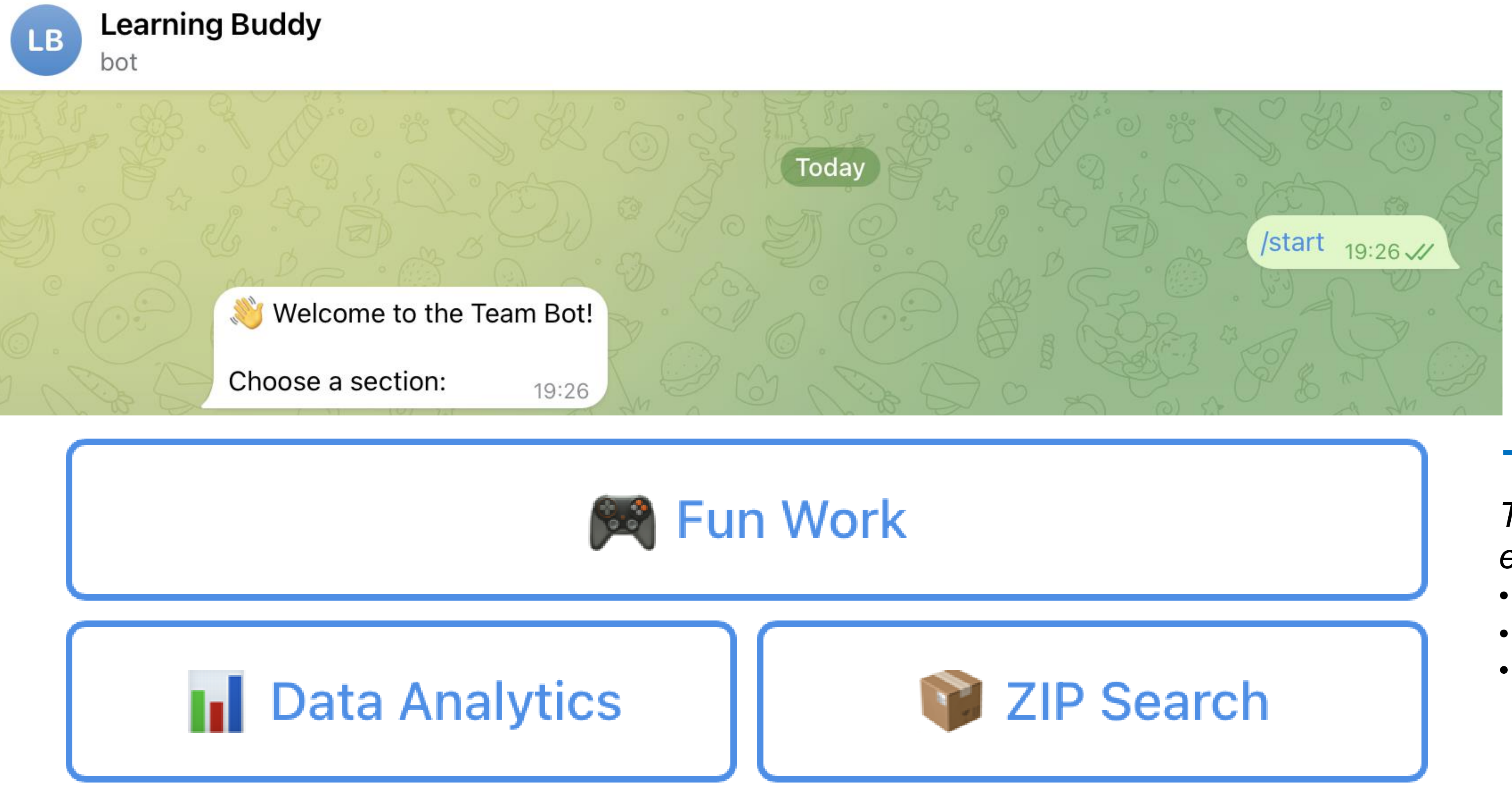
**Functionality:**
The bot uses a button-based menu to make interaction easier for users. The user can select options instead of typing long commands manually.

**Telegram bot name:**
Learning Buddy

**Telegram bot username:**
@MNU_learningbuddy_bot

# Main Menu and Navigation

The menu consists of three dimensions – each one designed for different purposes and consists of different functions.




*To begin working with bot – write down* ***'/start'***

*Three dimensions designed by each team participant:*

- ***Fun Work***
- ***Data Analytics with CSV File***
- ***ZIP Search***

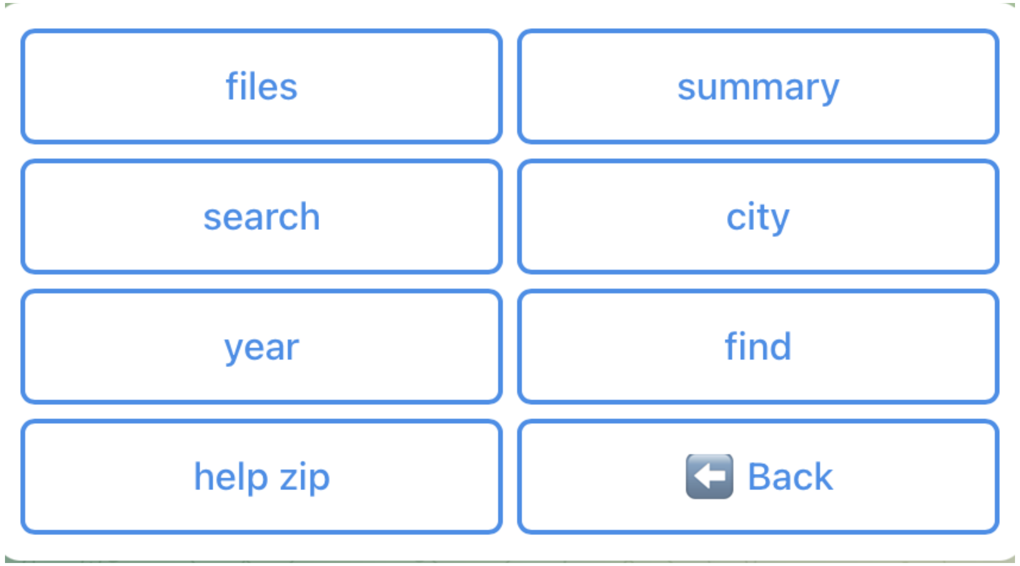
# Zip File Working Part

The user can upload a .zip archive. The bot saves the file and remembers it for the current user.

**Programming concepts used:**

- document handler
- file downloading
- zipfile
- dictionary user_zip_files
- try/except


*The ZIP file can be worked through different functions*



# Files Inside ZIP

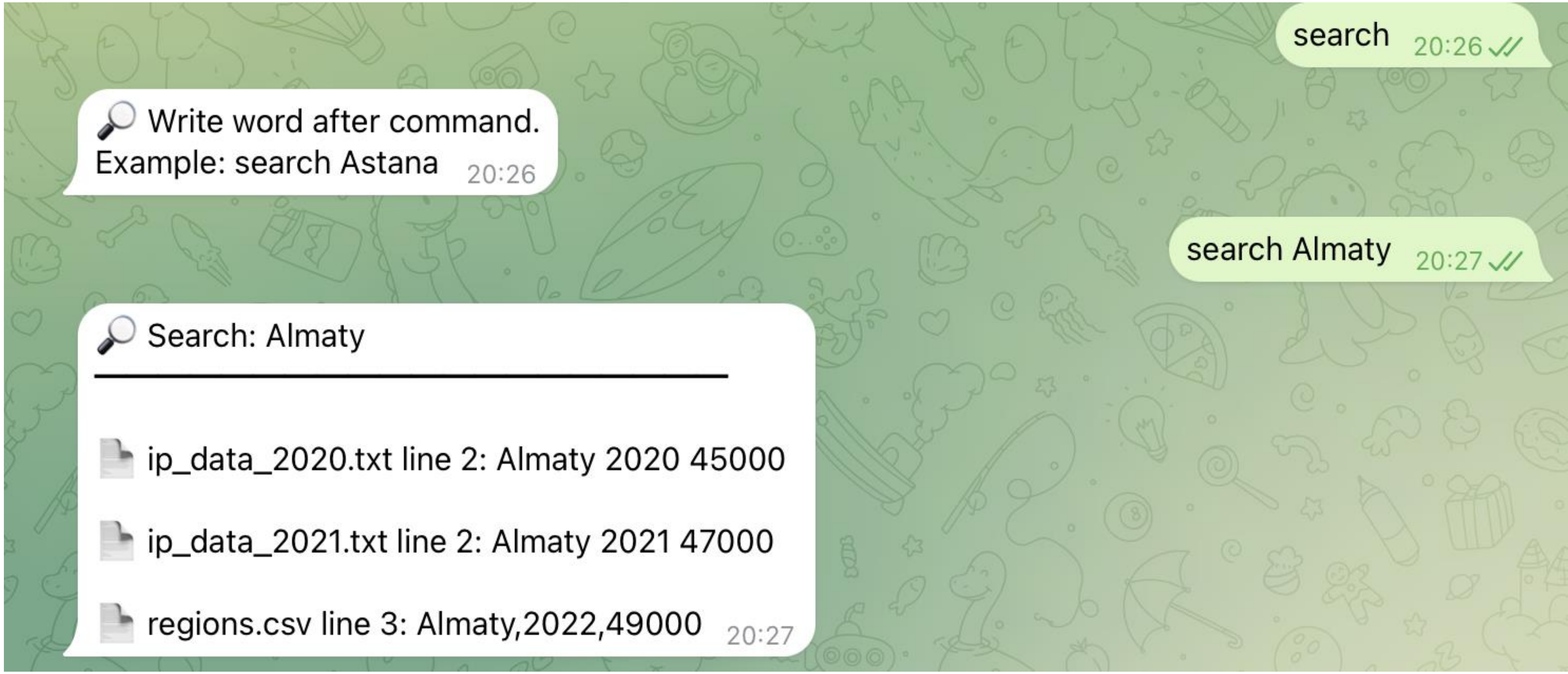
**Functionality:**
The bot opens the uploaded archive and displays all files stored inside it.

**Usability:**
The user can quickly check the structure of the uploaded ZIP file before searching inside it.

# Search Word in ZIP

**Functionality:**
The bot searches for a word in all .txt and .csv files inside the ZIP archive.

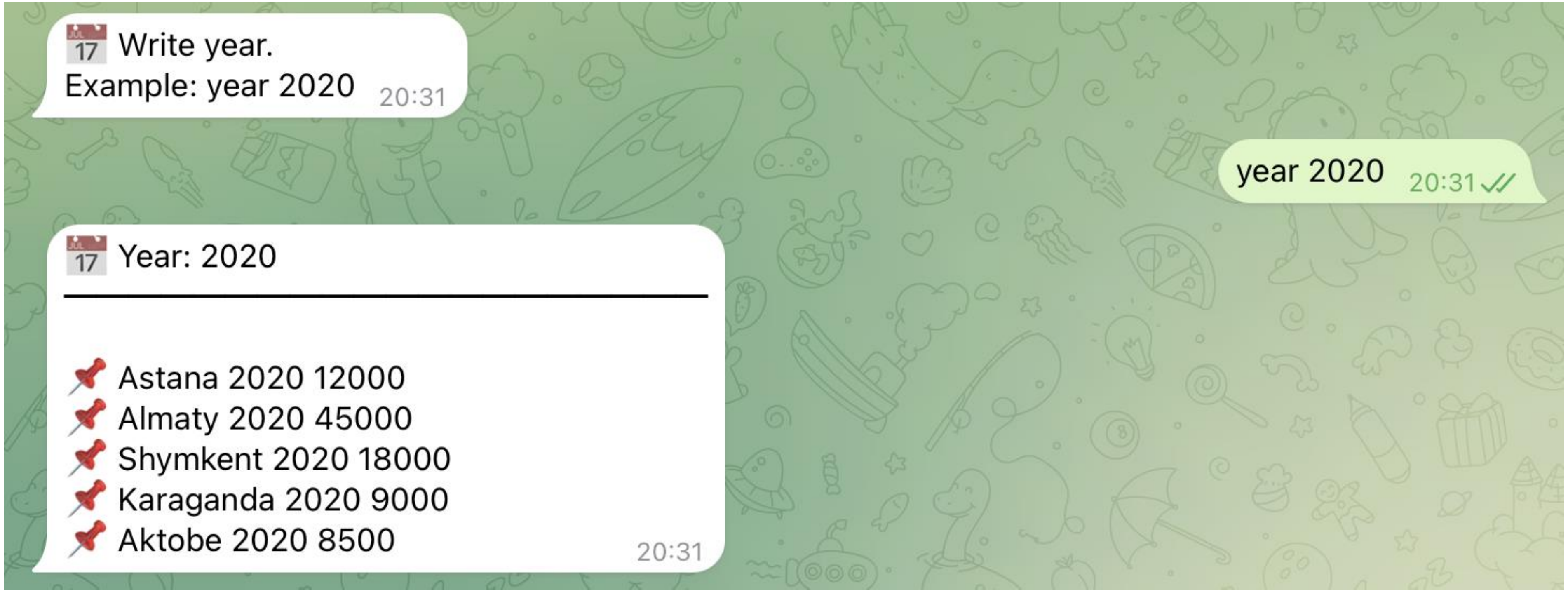
**Example:**
If the user writes search Astana, the bot finds all lines where this word appears.



# Search by Year

**Functionality:**
The bot searches all records connected with a selected year.

**Example:**
The command year 2020 returns all lines that contain data for 2020.



# Find by City and Year

**Functionality:**
This function combines two conditions: city and year.

**Example:**
The user writes find Astana 2021, and the bot returns only records where both values are found.

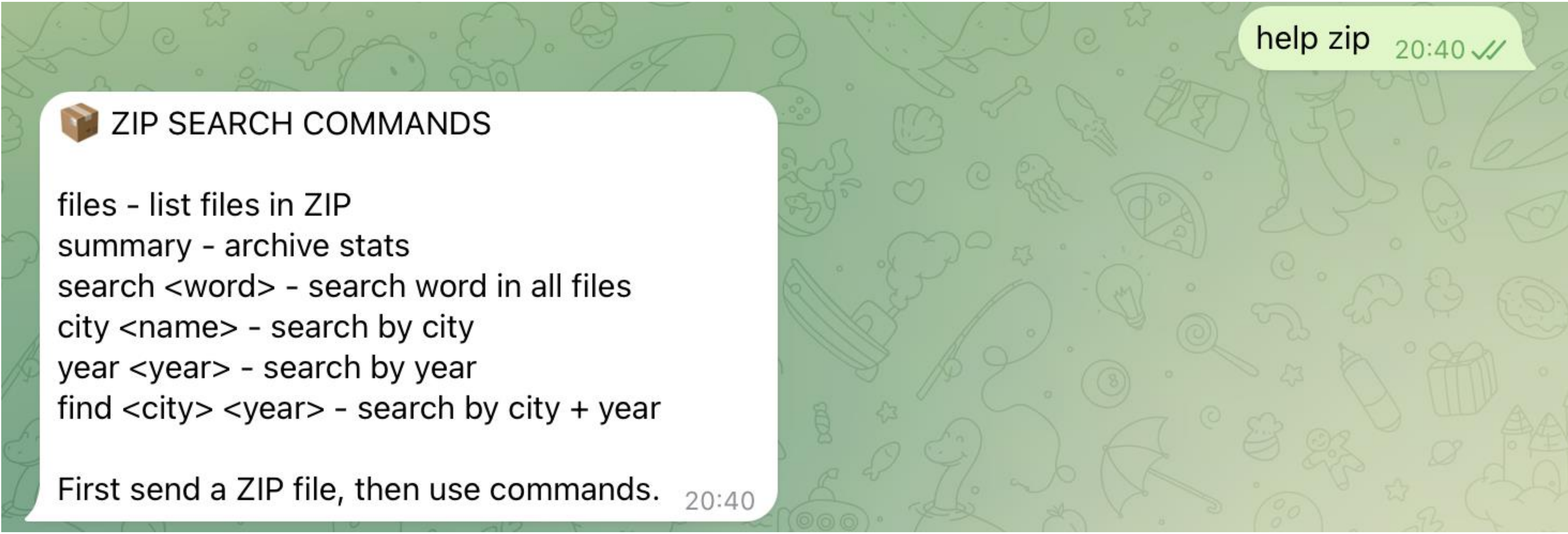
# ZIP Summary

**Functionality:**
The bot gives a short summary of the archive.

**It shows:**
- total number of files;
- number of .txt files;
- number of .csv files;
- total number of lines.

# ZIP help

If person needs any type of help in understanding of functions represented by ZIP dimension – ZIP help button shows commands description

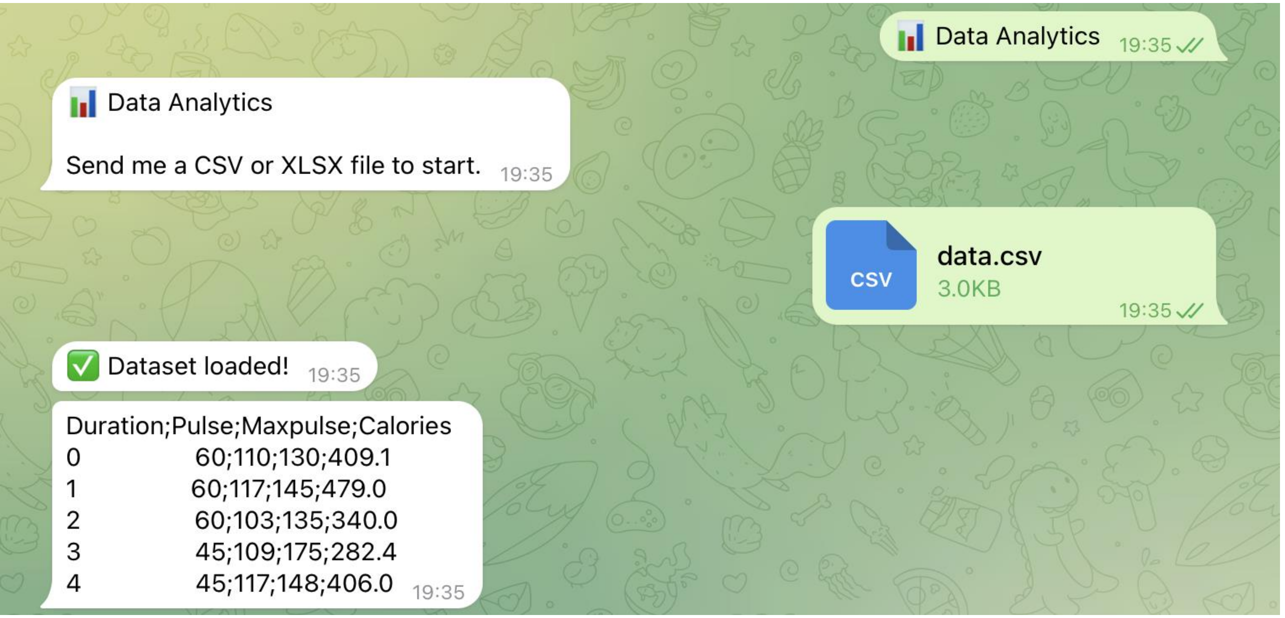


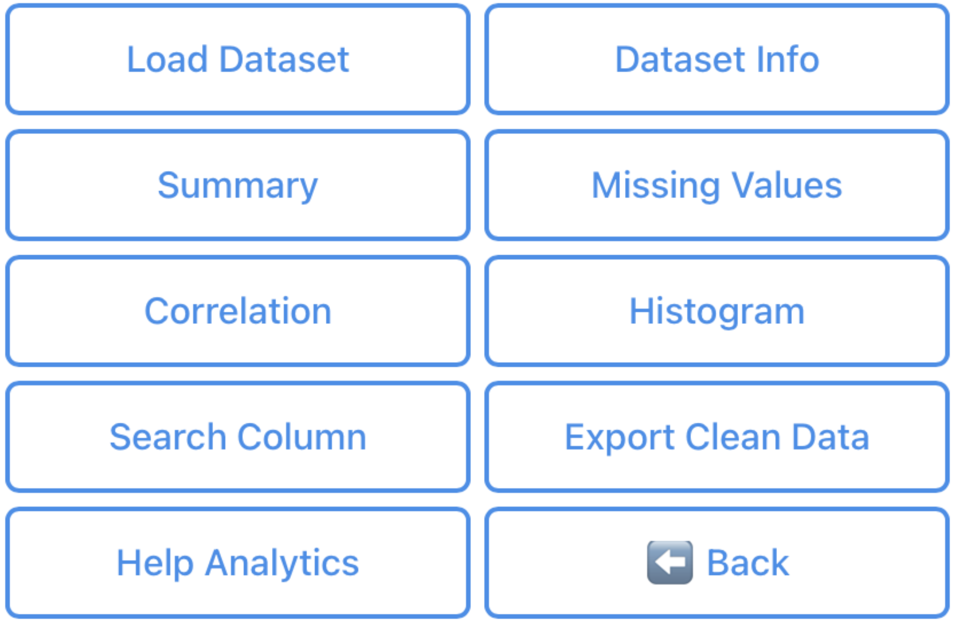
# Data Analytics with CSV

The user can upload a .csv archive. The bot saves the file and remembers it for the current user.

**Functionality:**
The bot provides a separate menu for data analysis. Instead of typing long commands, the user can press buttons.

*The CSV file can be worked through different functions*

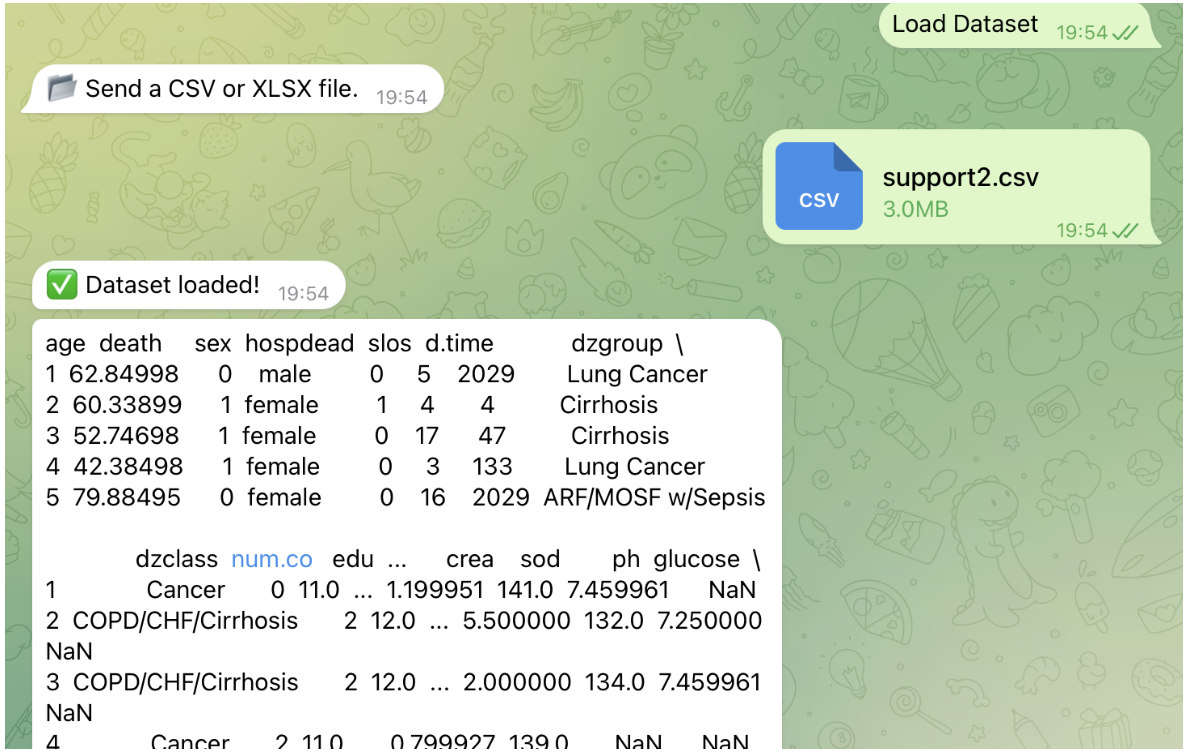


# Load Dataset

**Functionality:**
This function allows the user to upload a dataset file. After uploading, the bot reads the file and saves the dataset for the current user.

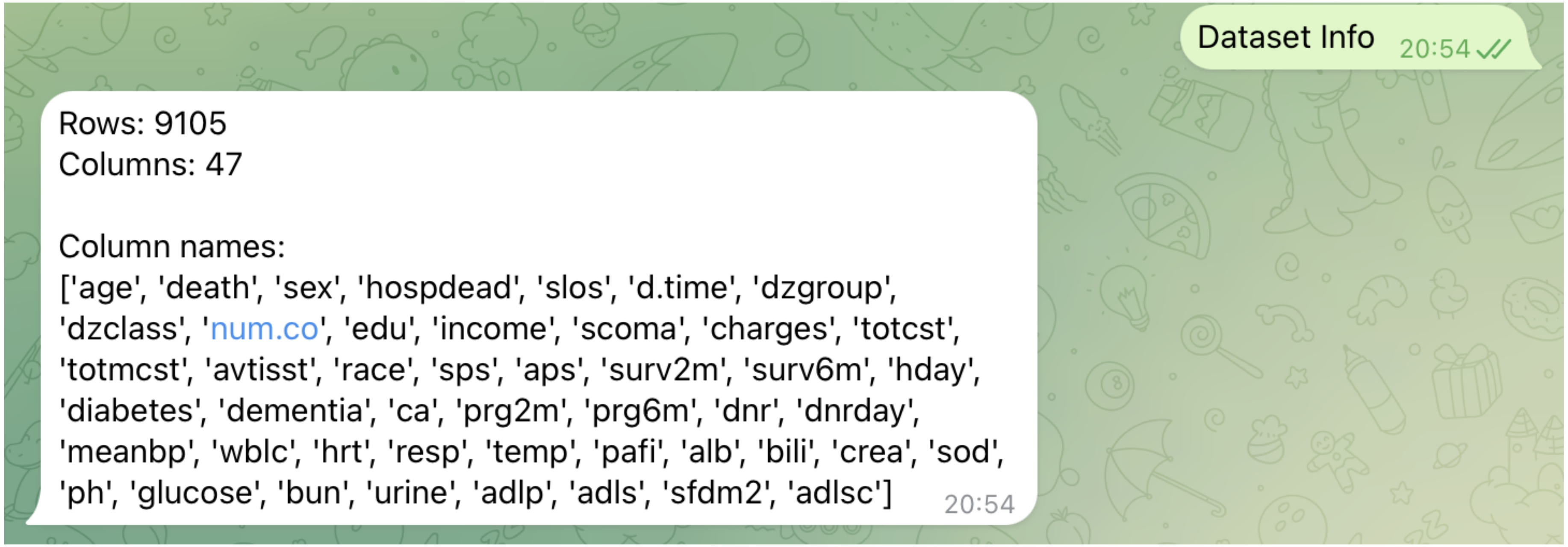
**Example:**
The user presses Load Dataset, then sends a .csv file. The bot reads it and shows that the dataset was loaded successfully.



# Dataset Info

**Functionality:**
This function shows basic information about the uploaded dataset.

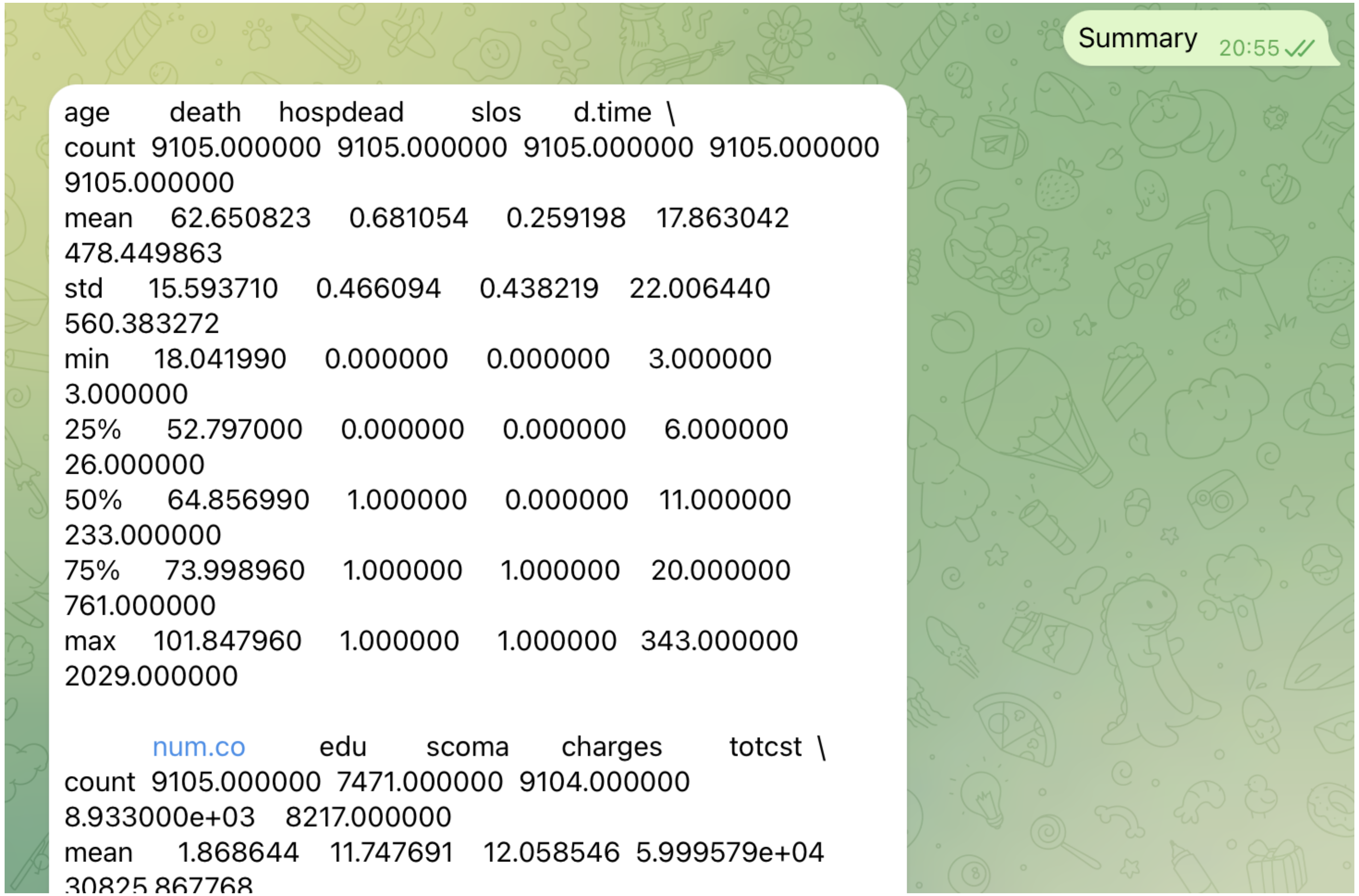
**The bot can show:**
- number of rows;
- number of columns;
- column names;



# Dataset Summary

**Functionality:**
This function shows basic statistical information about numeric columns in the dataset.

**The bot can show:**
- count;
- mean;
- standard deviation;
- minimum value;
- maximum value;
- quartiles.



# Dataset Missing Values

**Functionality:**
This function checks whether the dataset contains empty or missing values.

**The bot can show:**

- which columns have missing values;
- how many missing values each column has;
- whether the dataset is complete or not.

# Dataset Correlation

**Functionality:**
This function shows the relationship between numeric columns in the dataset.

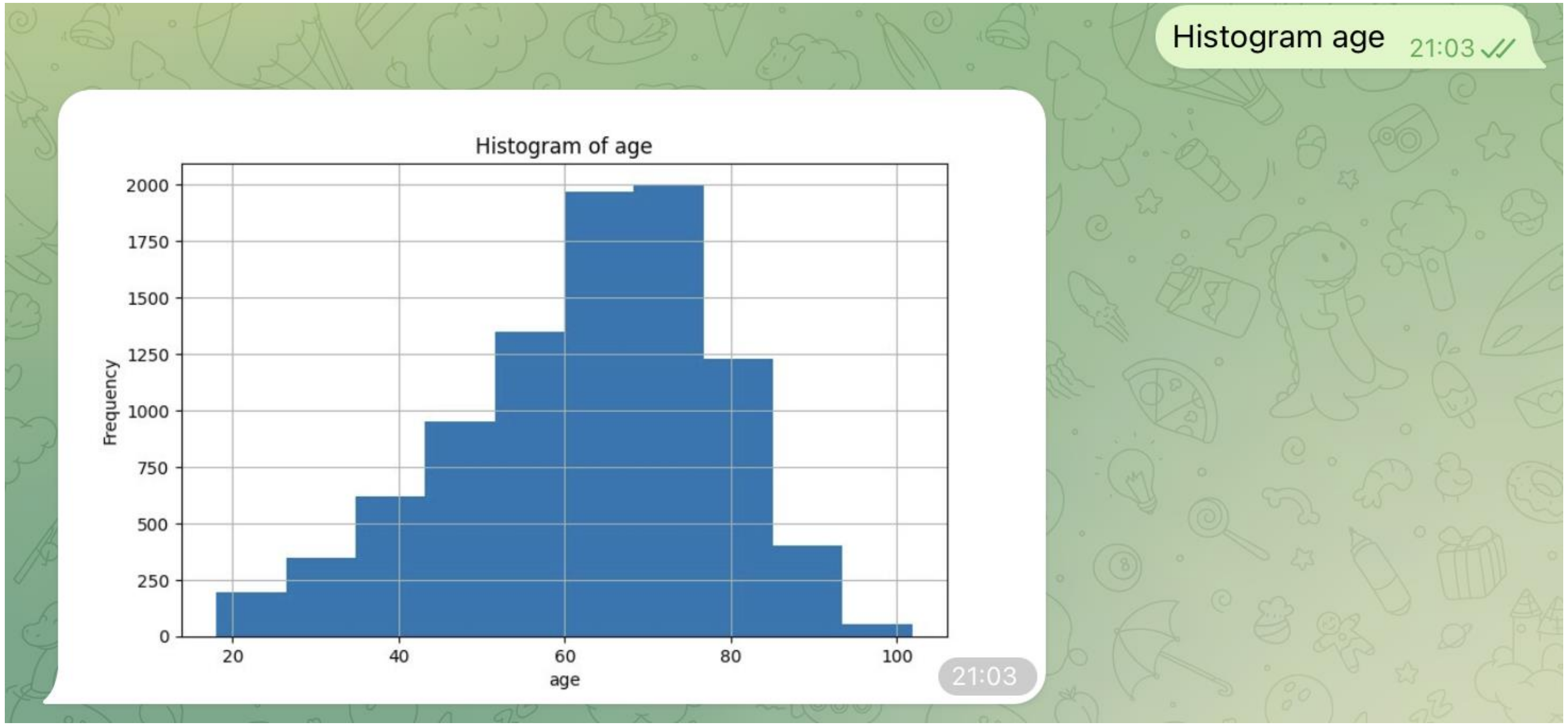
**What it does:**
The bot calculates correlation values between numeric columns and sends the result to the user.

**Why it is useful:**
Correlation helps understand whether two numeric variables move together.

# Dataset Histogram

**Functionality:**
This function creates a histogram for a selected numeric column.

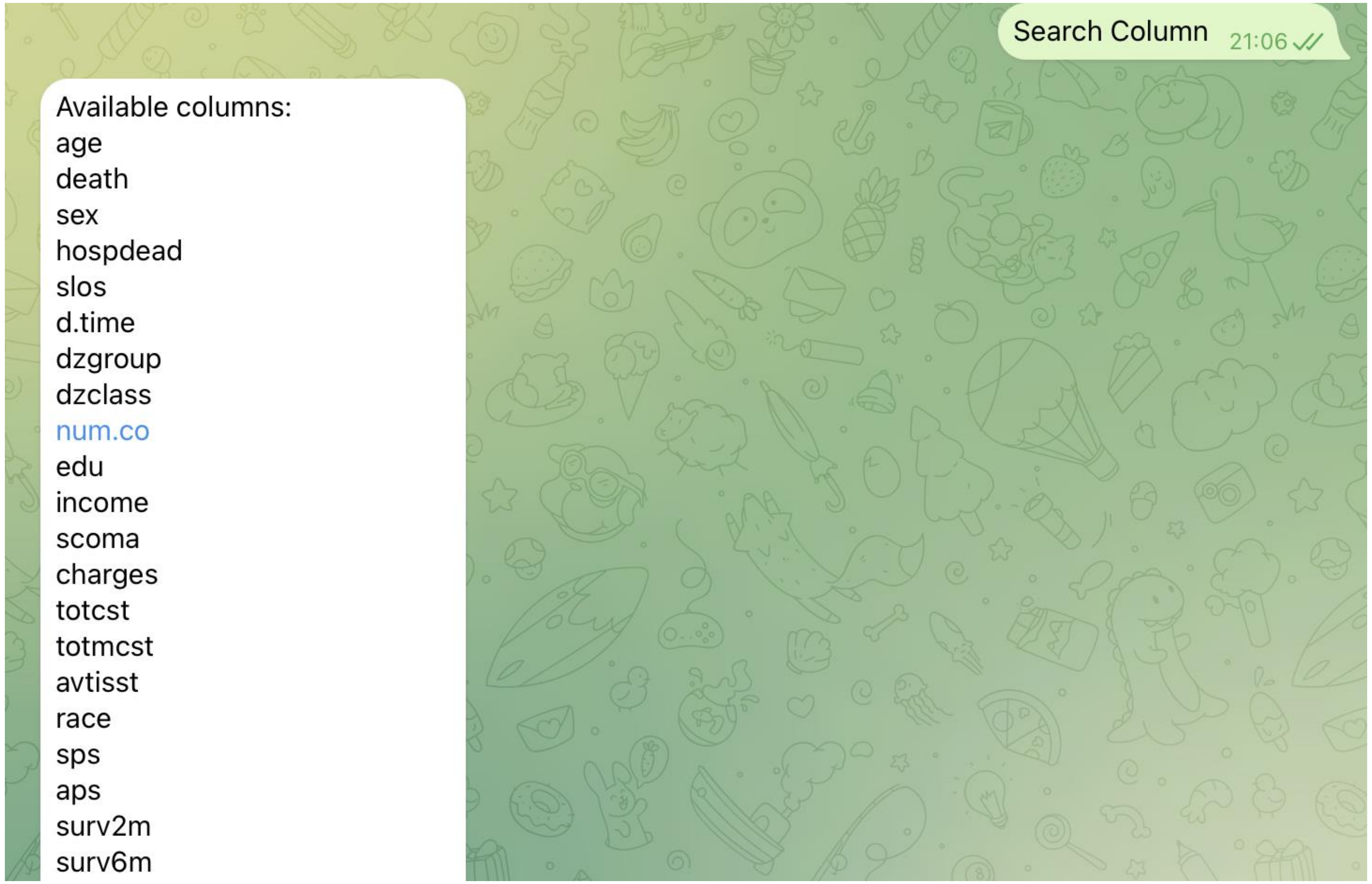
**Example:**
The user presses Histogram, then types a column name, for example:



# Search Column

**Functionality:**
This function allows the user to look for a column by avaliable columns list

**Example:**
If the dataset has columns:



# Export Clean Data

**Functionality:**
This function creates a cleaned version of the dataset and sends it back to the user as a file.

**What the bot can do:**
The bot can replace missing values or prepare a cleaner version of the dataset for download.

# Help Function

**Functionality:**
This function explains how to use the Data Analytics module.

**The help message includes:**
- available buttons;
- short description of each function;
- examples of user input;
- reminder to upload a dataset first.

**Why it is useful:**
It makes the bot more user-friendly and helps new users understand the available functions.

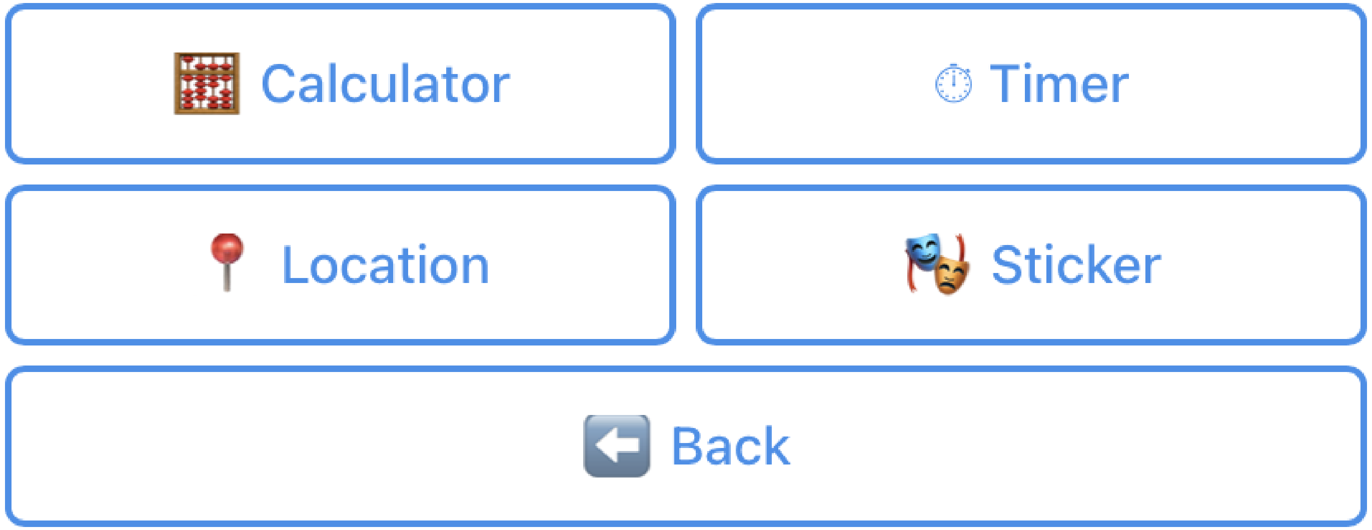
# Fun Work Part

This module was added to make the bot more interactive. It shows that Telegram bots can work not only with files and datasets, but also with user actions, buttons, stickers, timer functions, and location messages.

**Programming concepts used:**

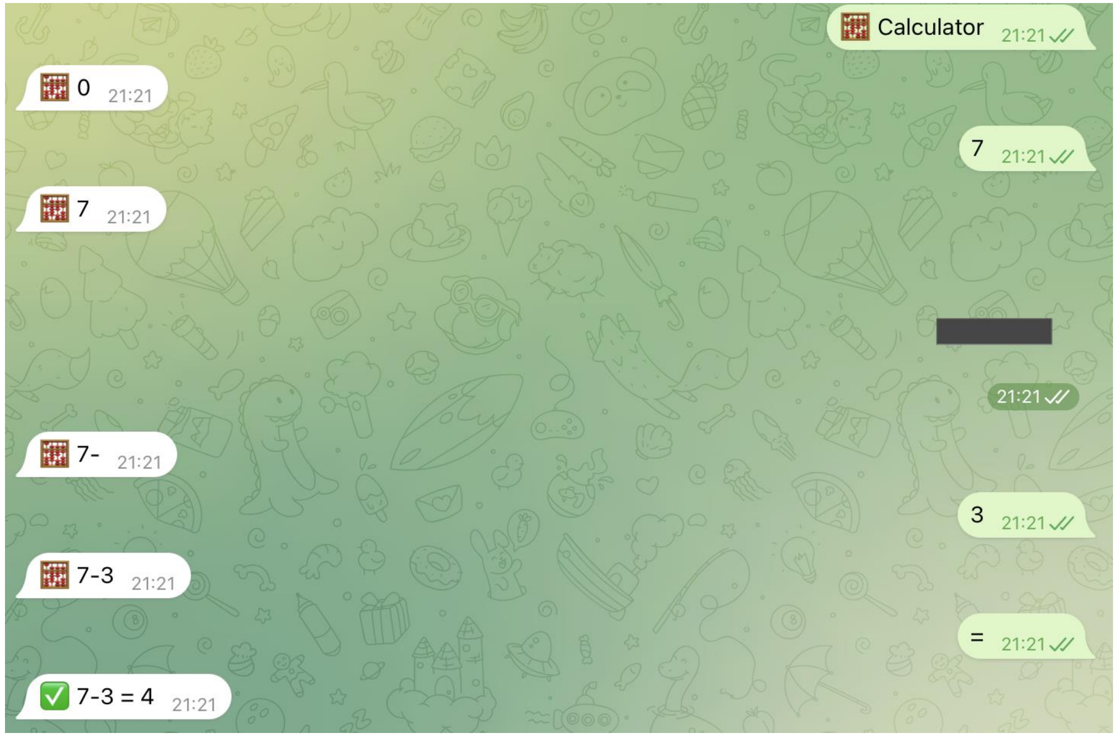
- Telegram message handlers
- Button menus
- User states
- Dictionaries
- Conditional statements
- Simple calculations
- Timer logic
- Location message handling
- Sticker sending
- Error handling with try/except

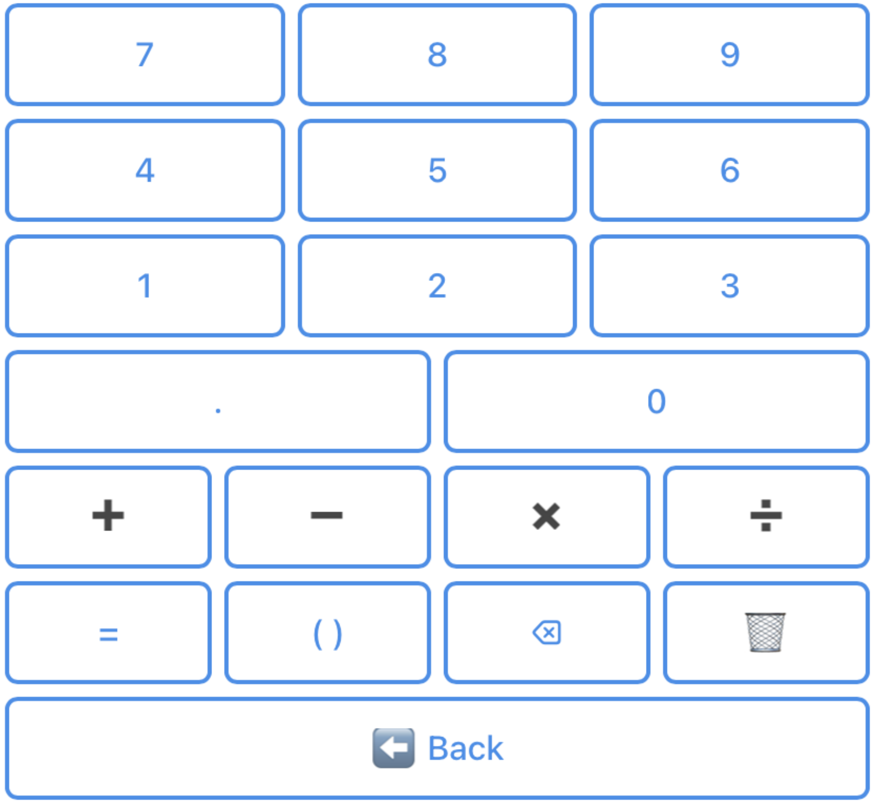
***The part consists of 4 internal functions such as:***



# Calculator

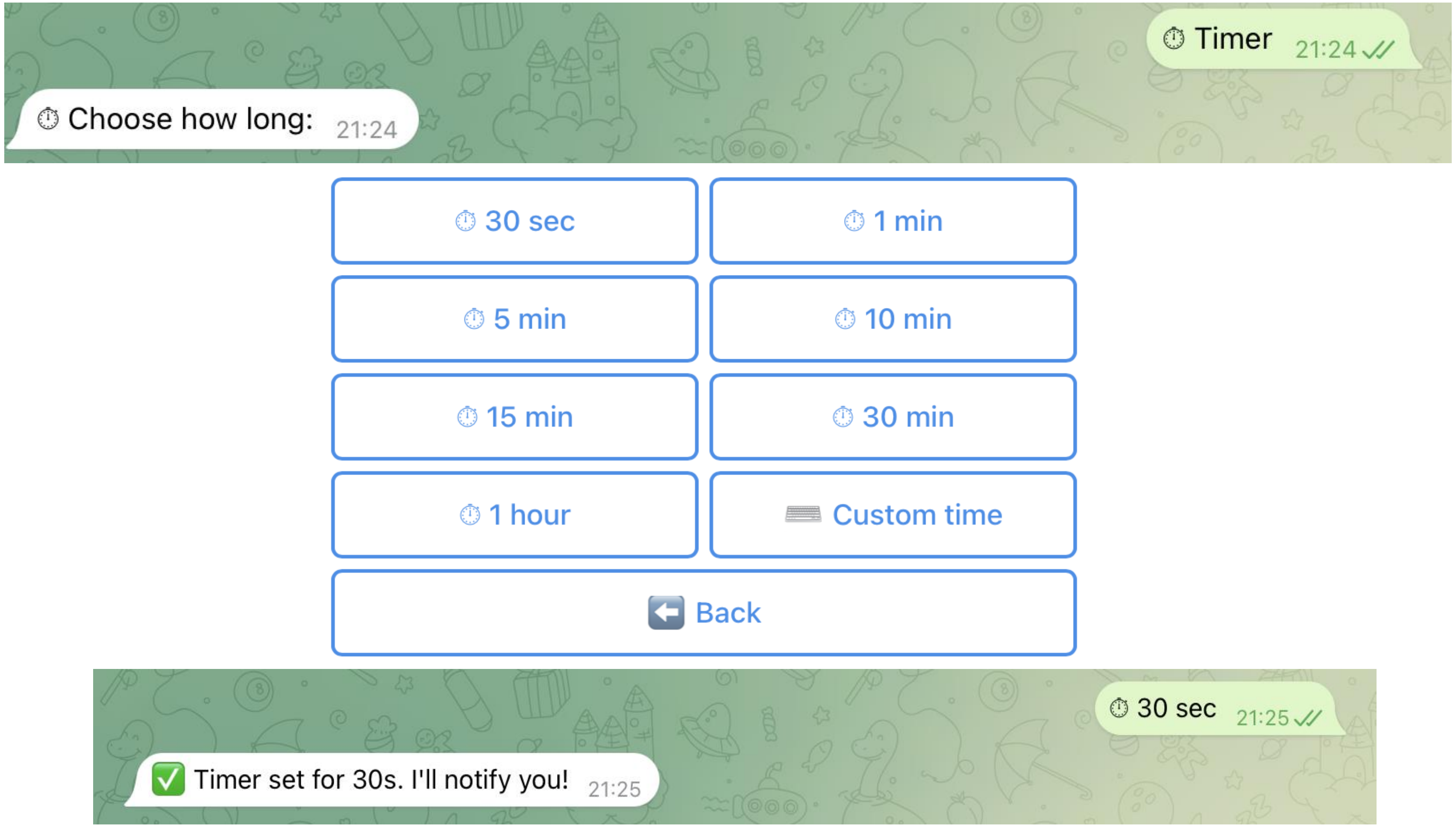
**Functionality:**
The calculator allows the user to enter a simple mathematical expression and receive the result directly in Telegram.



# Timer

**Functionality:**
The timer allows the user to set a simple reminder for a selected time.

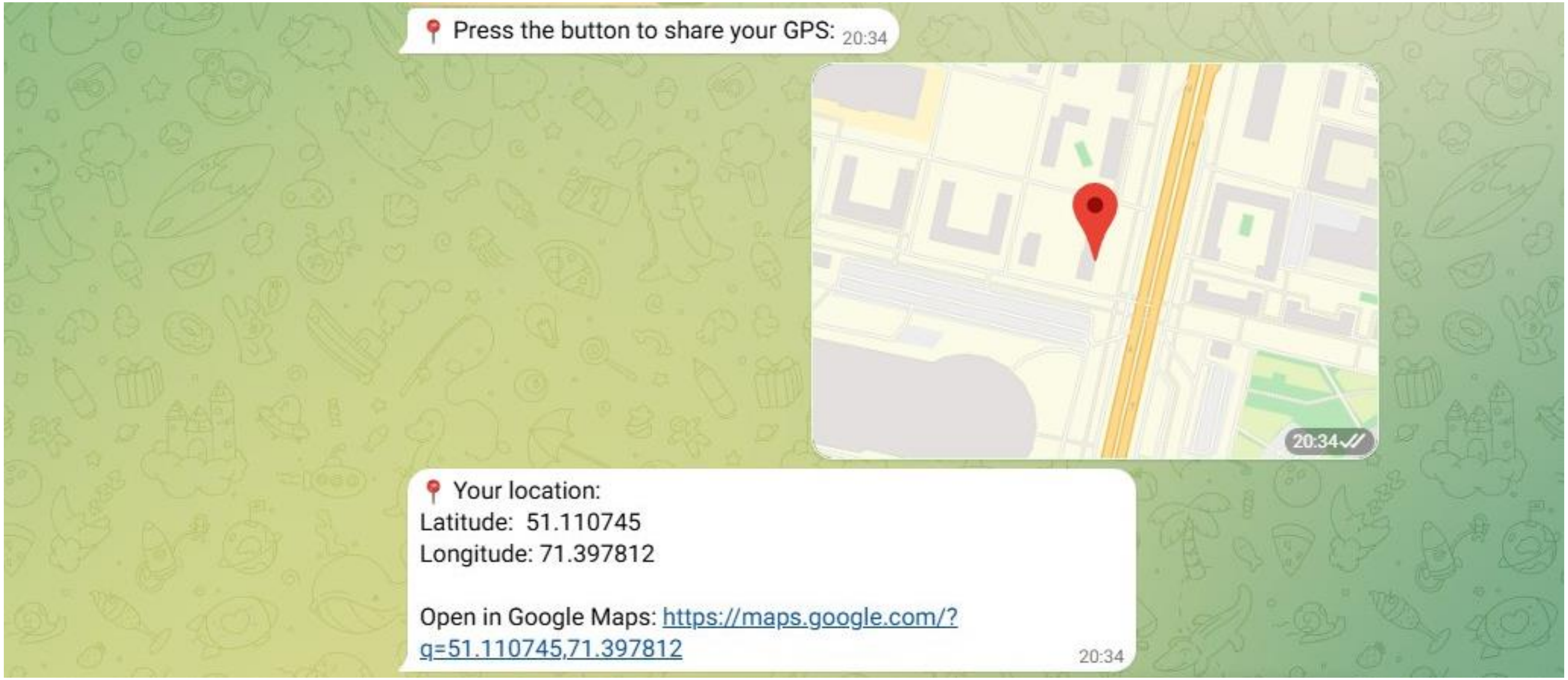


# Custom Time

**Functionality:**
The custom timer allows the user to enter their own number of seconds or minutes.

# Location Function

**Functionality:**
The location function allows the user to share GPS coordinates with the bot.
After receiving coordinates, the bot can create a Google Maps link using latitude and longitude.



# Sticker Function

**Functionality:**
The bot can send stickers when the user opens specific functions or performs certain actions.

**Example:**
When the user sets his timer, sticker will be sent immediately.

# Thanks for Attention

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